

Deadline: Dec. 19 2022

1. Show the relationship between monomer and repeating unit.
2. List a few examples for the configuration affecting polymer properties.
3. Calculate the number-average, weight-average molecular weight and PDI according to the following data.

Components	Weight Fractions	Average relative molecular weight
A	0.05	13000
B	0.09	20000
C	0.20	34000
D	0.18	45000
E	0.21	70000
F	0.13	112000
G	0.10	132000
H	0.04	156000

4. Tell the difference among step-growth polymerization, chain polymerization, condensation polymerization, and addition polymerization.
5. When will the gel point be reached in a system containing 2 mole RA₃ and 3 mole of R'B₂ provided the only reaction occurring is that between A- and B-groups?
6. To increase the MW in the process of polyester production, the byproduct (water) is required to be removed as much as possible from the polymerization system. Why? Can you explain it in your own words or show an example with simulation data.
4. Calculate the extent of reaction at which gelation occurs for the following mixtures:
 - a. Phthalic anhydride and glycerol in stoichiometric amounts
 - b. Phthalic anhydride and glycerol in the molar ratio 1.500:0.980
 - c. Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.990:0.002
 - d. Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.500:0.700